

Exploring the Impacts of AI Technologies on Digital Collage Creation

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Figure 1: Digital collage artworks created by participants.

Abstract

AI technology has been increasingly integrated into digital art-making tools to support artistic creation, such as automatically coloring sketches. Although research on the application of AI in art and creative practices is growing, its role in collage creation has seen limited research attention. To explore the potential of AI in digital collage creation, we followed a human-centered approach and conducted a co-design workshop with eight participants. Through this workshop, we examined how users interact with AI during the creation process, explored the impact of AI on their creation process and experience, and investigated their attitudes toward AI-assisted creation. Our findings offer valuable design implications for the future development of AI-assisted digital collage tools.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; • **Applied computing** → **Media arts**.

Keywords

Digital Collage, Generative AI, Generative Art

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1 Introduction

Collage is an art form that involves assembling various materials like paper, photographs, or other materials to form an artistic composition [1, 8, 24]. It requires little artistic skill or experience, making it inclusive and accessible to people of all ages to engage in the creative art-making process [17]. During the process of creation, collage may evoke memories, foster self-expression, and promote stress relief [12]. Thus, it has been adopted as a tool in art therapy and as a therapeutic practice that helps people manage stress [8, 28]. Traditional collage involves cutting and pasting physical materials, typically paper-based, onto a surface, while digital collage creation uses software tools to select and manipulate images on a screen, which offers greater flexibility for artistic creation and expression. These digital tools expand material options and allow for more flexible layer adjustments [16, 24]. However, the process still involves multiple steps—such as selection, editing, and composition—that require significant effort. For example, it usually takes time to find materials that match user needs from a large pool, and arranging and editing elements also demands high user effort.

In the broader field of art creation, digital tools have been extensively used to support various forms of art-making such as painting

and drawing (e.g., ProCreate¹). Some of them have incorporated advanced features powered by AI technologies to support art creation, including content recommendation, content understanding, and image generation [5–7]. While AI has been increasingly studied in the field of art and creativity, most existing research focuses on applications in painting and drawing, such as automatically coloring sketches [23] or generating drawings based on sketches [29]. However, collage creation differs from these tasks, as it focuses on arranging and integrating existing visual elements rather than creating entirely new images. This distinction makes it essential to investigate AI's role in collage creation separately. Considering the complexity of the multi-step collage creation process, we seek to explore whether AI can provide support throughout the creation process. Additionally, we are also interested in exploring the potential impact of AI on the digital collage creation experience and its associated therapeutic benefit.

For this investigation, following a user-centered approach, we conducted a design workshop with eight participants to explore the use of various AI tools (e.g., ChatGPT², OpenArt³) in the process of digital collage creation. Through the workshop together with focus group interviews, we analyzed how participants used the provided AI tools across four stages of the collage creation process and their impact on both the creation process and their overall experience. The results revealed that AI could enhance efficiency, creativity, and user experience in digital collage making, but maintaining user autonomy throughout the creation is essential. Additionally, participants shared their expectations for AI integration in digital collage creation, such as providing real-time support as a companion and offering proactive guidance to reduce user input effort. This study makes three key contributions: 1) it presents insights from a design workshop that captures the practical use of AI tools in digital collage creation; 2) it examines how these AI tools impact users' creation efficiency and experience; and 3) it offers design implications for developing AI-based digital collage tools based on the findings.

2 Related Work

2.1 Collage Creation

In the field of art, collage is an art form that allows creators to explore their creativity [1]. Collage comes from the French word *coller*, which means “to glue”. A collage can be completed simply by selecting, cutting, composing, and then gluing materials onto a surface [26]. Various materials can be used for collage, such as magazine images, photographs, or even fabric [1, 15]. The immersive process of assembling diverse materials into unique compositions enables individuals to explore their creativity while fostering self-awareness and self-expression, making collage creation an effective tool for emotional expression and stress reduction [12]. Therefore, it has also been widely used as a therapeutic method in the context of art therapy, helping to explore people's underlying thoughts and feelings and foster their self-expression [28].

Moreover, the advent of digital collage tools has expanded the creative possibilities of collage making. Although digital collages

may lack the sensory experience of traditional art and pose challenges for those unfamiliar with digital technology [8], it overcomes the limited selection of materials in traditional form, allows for flexible composition, and makes it easier to save, share, and review works, thereby offering creators a wider scope for creative expression [4, 16, 21, 24]. Whether traditional or digital, collage allows for self-expression and allows others to gain a deeper understanding [22]. Currently, the use of digital collage in fields like teaching [25] and art therapy [16] underscores its value as a multi-functional tool.

2.2 AI Technology for Digital Art Creation

In the field of visual art creation, AI technologies have had a profound impact on the creation process. AI assistance not only simplifies the painting process but also provides creative inspiration. For example, a web application called Petalica⁴ can automatically color users' sketches, while an interactive system named Creative Sketching Partner can generate inspiring new sketches based on drawn drafts [13]. Furthermore, previous research also suggests that AI may have a positive therapeutic effect on artistic creation, e.g., using AI painting software can positively impact users' emotions [31]. Studies have also explored the use of AI for the advancement of digital art therapy [9, 14]; for instance, AlphaDARF, an AI-driven expert support system, helped art therapists and psychologists with large-scale automated assessments of clients' psychological states through drawing analysis [14].

AI-integrated tools like Canva⁵ can be used to create digital collages [10]. These tools leverage AI features to assist users in their creative work, such as generating desired materials and automatically removing backgrounds. However, it remains unclear whether and how the utilization of AI can influence the creation process and creation experience, as well as people's attitudes toward using AI in digital collage creation. Therefore, this study aims to explore these aspects, gaining insights for the future design and improvement of digital collage creation tools.

3 Methodology

The study was conducted through a design workshop involving eight college students. Participants were asked to explore and use a selection of AI tools to create a digital collage. Following the collage-making activity, a semi-structured group interview was conducted to explore how the use of AI influenced their creation process and overall experience.

3.1 Participants

We recruited eight young participants, including three males and five females, aged 19 to 22 (two participants aged 22, one aged 21, two aged 20, and three aged 19). Of the recruited participants, five had previous experience with traditional collage-making, while the other three participants had no prior experience in this area. Before the workshop, six participants were somewhat or very familiar with AI tools, while two had limited experience. Most of our participants were familiar with LLM-based tools like ChatGPT and DeepSeek [19], while some had experience using creative

¹<https://procreate.com>

²<https://chatgpt.com/>

³<https://openart.ai/>

⁴https://petalica.com/index_en.html

⁵<https://www.canva.com/>

Table 1: AI Experience Materials

Step	Techniques ID	AI Techniques	Task Support	Tools
Selecting	TG	Text Generation	Provide ideas or inspiration to help users select themes and materials.	ChatGPT
	IG	Image Generation	Generate a new image based on user-specific needs, rather than sourcing from existing resources.	DALL-E 2 ⁶
	ISR	Image Search & Recommendations	Recommend images based on the input queries to help discover materials from large databases.	Google ⁷ , pexels ⁸
Editing	IE	Image Editing	Modify images such as removing background, correcting color, or enhancing image quality.	OpenART , Picsart ⁹
	IG	Image Generation	Generate a new image based on the original image to support image extension, style transfer, and other tasks.	OpenART
	IU	Image Understanding	Segment and crop images based on content understanding.	Segment Anything
Composing	ICO	Image Composition Optimization	Adjust the material layout and optimize the composition automatically.	PowerPoint ¹⁰
Finalizing	IU	Image Understanding	Generate textual descriptions based on the understanding of collage content.	DALL-E 2
	TG	Text Generation	Generate text content to help add captions and explanations to collages.	ChatGPT

AI applications such as Midjourney¹¹ for generating images and Suno AI¹² for creating music. The university’s Institutional Review Board approved the study protocol based on ethical considerations (FWA00021580). All participants provided informed consent, including authorization to use their artwork (see Figure 1) and photos. To ensure privacy, pseudonyms were assigned, and faces in photos were blurred. Participants were compensated with a 150 RMB (approximately 22 USD) gift card for their participation.

3.2 Materials

Participants were asked to prepare for the workshop by reviewing and testing our experimental digital collage tool, as well as the AI tools they might use for creating collages. This preparation ensured a smoother creation process during the workshop, minimizing confusion and technical barriers, and enabling participants to focus more fully on the collage-making experience. The day before the workshop, we provided participants with a user guide for the digital collage tool FreeMix¹³, which instructed them to use this tool to create digital collages. We selected FreeMix as the experimental tool for our study due to its clean interface, user-friendly features, and support in key steps of collage-making, such as manually cutting materials with a mouse to mimic the scissor process.

Since the digital collage tool FreeMix lacks built-in AI features, we selected AI techniques based on their potential to support tasks across the four key steps of collage creation [26], as outlined below:

- **Selecting:** Define a theme and choose relevant materials, including images, text, drawings, and other elements.

- **Editing:** Edit the materials based on personal needs, including cropping, rotating, resizing, color correction, duplicating, erasing, and flipping.
- **Composing:** Drag and arrange the edited materials on the canvas, and optimize the composition by adjusting the layer order.
- **Finalizing:** Add text descriptions or drawings to refine the details. Once complete, the collage work can be saved, exported, and shared.

We structured the introduction and usage guidelines for each AI technique, along with the tasks they support in the collage creation process, into a user guide for using these AI tools during the workshop. Table 1 summarizes the AI techniques chosen for each step of creation, highlighting their relevant features for the tasks and representative tools. For example, in the “editing” step, Segment Anything¹⁴ was expected to help users precisely segment objects within an image and automatically crop them, thereby reducing the difficulty of manual cropping. In the following sections, we refer to these AI techniques using the IDs listed in Table 1.

3.3 Workshop

At the start of the workshop, the eight participants were randomly divided into two gender-balanced groups: P1-4 in the first group, and P5-8 in the second. Each group was assigned a facilitator to guide the discussion, promote engagement, and observe the creation process. An ice-breaking activity followed, in which participants introduced themselves to foster open dialogue.

Digital Collage Creation. In this phase, we introduced the basic concepts of traditional collage, the creation process, and its benefits to mental well-being. Additionally, we provided an overview of digital collage tools, highlighted the advantages of digital collage

¹¹<https://www.midjourney.com/home>

¹²<https://suno.com/home>

¹³<https://www.freemix.com/>

¹⁴<https://segment-anything.com/>

over traditional collage, and the potential use of AI in supporting digital collage creation. In addition, we explained the selection of FreeMix as the experimental platform for the workshop. We allocated 40 minutes for participants to create their digital collages and encouraged them to freely choose a theme for artistic expression and use the provided AI tools at their discretion.

Focus Group Interview. During the focus group interview, the facilitator asked participants to reflect on their collage-making experience and share their thoughts one by one. Guided by prepared interview questions, participants engaged in in-depth discussions about each step of the creation process. Specifically, they were asked to: 1) describe which AI tools they used and how they applied them, 2) discuss the impact of these tools on their creation process and overall experience, and 3) share their expectations for how AI could better support digital collage creation, as well as any concerns they had about using AI in this context.

3.4 Data Analysis

The data analysis was conducted using an inductive thematic analysis approach [3] with the aid of ATLAS.ti for organizing and coding qualitative data. Two authors independently performed open coding on 30% of the focus group interview data for each question, identifying preliminary themes. They then discussed their codes to refine them and resolve discrepancies, resulting in the development of a codebook to ensure consistency and rigor in the analysis. Using the finalized codebook, the first author completed the coding of the remaining data.

4 Findings

4.1 AI Tool Usage

AI technologies provided multilayered support throughout the digital collage-making process, and each tool offered unique functionalities at different steps:

Selecting. **ISR** was the most widely used and helped all participants acquire creative materials in the initial phase, while **IG** played a crucial role in the transformation of textual descriptions into visual content. **TG** supported theme refinement and material selection (P2, P3). As P3 noted: “I wanted to express, in just a few sentences, how feminism is rising while decayed things sink into the sunset and new things emerge with the sunrise. So, I asked ChatGPT how to concisely convey this theme.”

Editing. In this step, **IE** was used to remove unwanted elements from images and remove and modify image backgrounds (P4, P6, P7). **IG** helped P5 enrich the content of images. Some participants also used Segment Anything from **IU** to eliminate redundant content (P1) and extract specific parts from complex images (P2).

Composing. We did not find an existing tool specifically for **ICO**, so we demonstrated the function of optimizing visual composition using the “Designer” feature in PowerPoint. While none of the participants tried this AI technique, some noted they might not need it for composition, as arranging elements often reflects personal feelings and emotions that AI cannot fully understand. However, P1 mentioned that **ICO** could be helpful in arranging elements according to their intentions, particularly when the arrangement requires precise operations, such as positioning decorations into a heart shape.

Finalizing. Regarding **IU**, participants used it to analyze and interpret the content of their collages, gaining a deeper understanding of their work (P7, P8). In addition, **TG** helped P2 create a textual description related to the theme and generate a short phrase for P3 to incorporate into the collage as an artistic embellishment.

4.2 Impact on Creation Process

Through the discussion, we identified both positive impacts and challenges of using AI technology in the creation process.

On the positive side, participants highlighted several benefits. Many stated that AI tools not only helped *reduce the time spent acquiring materials*, but also *generated images that met specific needs, without being constrained by existing resources*. For instance, P1 noted that **IG** helped him generate images of future scenarios and past memories, which are difficult to find in existing resources. Additionally, AI tools *provided creative inspiration and suggestions*. For example, P3 shared that AI helped them determine a theme, and P2 received helpful material suggestions through their interaction with ChatGPT in **TG**. AI also *reduced the difficulty of image editing*, with participants praising its effectiveness for tasks like background removal and modification (P8), image segmentation and cropping (P1, P2), and removing unwanted elements from images (P7), which previously required significant manual effort. Lastly, AI tools were seen as enhancing the *aesthetic arrangement of materials*. P8 noted that **ICO** could simplify the process of adjusting layers and designing layout frameworks, transforming disorganized compositions into visually appealing designs with minimal effort.

On the negative side, participants identified several challenges. For AI tools requiring text descriptions, they felt that *the cognitive load from creating suitable prompts could impact the collage-making process*. P1 noted a challenge with **IG**: “Users typically use short prompts to generate images, but this often does not produce good results. Writing precise sentences takes practice... It requires effort from the average user to come up with a good prompt.” Meanwhile, P6 and P8 mentioned that such cognitive load sometimes distracted them from their creative focus when using **ISR**. When discussing **ICO**, several participants expressed concerns that *AI-driven composition adjustments could interfere with their creative ideas*. P4 felt that such tools might ignore the personal expressions embedded in their designs, while P1 emphasized that every element on the canvas carried meaningful significance, and their relative positioning was crucial to their creative vision. Consequently, these tools could constrain their artistic expression. Additionally, P7, who used **IU** for work analysis, remarked that *understanding the content of completed works had minimal impact on the creation process*, as their artwork was already finished.

4.3 Impact on Creation Experience

We asked participants to reflect on each step of the collage-making process and their experiences with the AI tools, categorizing these experiences as either positive or negative.

Positive Experiences: Some participants reported experiencing *serendipity* while using AI tools. For example, P5 noted that the material recommendations provided by **ISR** were occasionally refreshingly unexpected. Similarly, P2 found that **TG**’s material recommendations not only aligned with their theme but also offered

unexpected inspiration, significantly boosting their creative motivation. Many participants also expressed a sense of *satisfaction* when AI tools met their expectations. Tools used during the Editing phase and **ISR** often delivered satisfactory results, leading to positive emotions and a joyful mood. Moreover, some participants experienced positive *emotional resonance* when interacting with AI tools. For instance, P3 shared a moment of personal connection when **TG** helped them select a theme:

“The generated result closely mirrored my life. Like in the output, the unhappy moments in my life fade away with the sunset, while the joyful and beautiful experiences rise with the sunrise. So I narrowed the theme down to my life, and my theme became ‘It’s my life.’”
(P3)

Negative Experiences: Despite the benefits, participants also encountered negative experiences with AI tools. One common issue was *dissatisfaction* when AI failed to deliver results that met their expectations. This led to negative emotions such as disappointment (P7), confusion (P4), and even disgust (P8). Participants also experienced *frustration* during the creation process, particularly when using **IG**. Inappropriate or imprecise prompt inputs often caused frustration and hindered the creative flow. Furthermore, some participants described a sense of *detachment* when engaging with AI-generated content. P6, who was particularly sensitive to language, felt that the text produced by **TG** lacked warmth and a human touch, describing it as overly marked by AI. P3 echoed this sentiment, stating that when **TG** provided inspiration through a Q&A format, the responses felt cold and impersonal.

4.4 Expectations and Concerns

In addition to improving the performance of AI tools, participants also highlighted several key expectations for integrating AI into digital collage tools. *First*, there was a strong desire for real-time conversational support as a creative companion. For example, P3, who felt a sense of resonance during the creation process, envisioned an AI companion that could provide emotional support and positive feedback in real-time. P8 emphasized the importance of conversational interactions that could adapt to their speaking style, creating a more natural and comfortable user experience. *Second*, participants hoped to reduce the cognitive load associated with crafting prompts. They proposed specific features to achieve this. P1 suggested that the AI could offer guidance by predicting the next part of a text based on what the user has already entered. Similarly, P2 proposed that the AI could generate complete sentences from a few provided keywords, streamlining the prompt-creation process. *Third*, visual optimization emerged as a key expectation, with participants focusing on two main areas: enhancing the visual attributes of individual materials and improving overall collage composition. P2 hoped for AI support in refining material properties, such as color and clarity, while P8 suggested that AI could assist by automatically arranging cluttered images on the canvas based on their content, resulting in a more aesthetically pleasing design. *Lastly*, participants emphasized the importance of maintaining user control for refining and adjusting details. P1 remarked, “AI can help create something from nothing, but achieving refinement and perfection often requires human intervention.” This sentiment

was echoed by others, who agreed that while AI can generate initial ideas, human input is essential for fine-tuning the output and ensuring the final result aligns with their creative vision.

In addition to these expectations, participants also voiced concerns about AI’s role in the creation process. The most significant concern was that AI might dilute original thinking. For instance, P7 worried that allowing AI to automatically arrange collage elements could interfere with their creative intent. Similarly, P3 highlighted the risk of AI-generated materials undermining their ability to create independently, as the tool’s interpretation of a prompt might overshadow their emotions to be expressed in the collage.

5 Discussion

Based on our findings, we identified several key aspects that should be considered when incorporating AI into digital collage creation, and we have provided concrete design implications below.

AI should prioritize user autonomy in digital collage creation by remaining entirely optional and empowering users to lead the creation process. Ensuring user autonomy is crucial in AI-assisted digital collage creation. Collage creation is a creative practice that highly relates to people’s inner feelings and self-expression [28], so it is crucial for users to lead the entire creation process. In our study, participants expressed varying needs for control, depending on the different stages of the creation process. For instance, during the selection phase, they may sometimes want to choose images or materials based on their thoughts in mind, but at times, they may appreciate suggestions to offer them inspiration. Furthermore, in the composing stage, some participants viewed it as a way to reflect their values and thoughts, as also suggested in the literature of collage therapy [2], so they would like to maintain the whole control and make their own arrangements. This insight suggests that AI should act as a meaningful collaborator, helping users push creative boundaries rather than taking over the creation process [11]. We suggest that AI should be entirely optional in digital collage creation, allowing users to decide whether and how to apply AI assistance based on their personal preferences.

The interactivity of AI would be favored, as people tend to prefer an interactive approach in the process of creating digital collages. Our study found that when engaging in conversations with AI tools such as ChatGPT, our participants not only gained creative inspiration and suggestions but also experienced a sense of resonance. This sense of resonance may stem from our participants perceiving AI as capable of understanding their intentions and providing the relevant feedback they expect [18, 20]. Based on this, participants expressed a desire for a real-time support system that offers conversational interaction as a companion, and they wanted to engage in dialogue with this system at any stage of creation. We think that such a companion system can be integrated with various AI technologies. For example, our participants hoped that AI could provide guidance during text input, while the companion system would actively offer prompts through conversational interactions [30]. This approach would not only minimize the effort required to formulate suitable text prompts but also deliver an engaging and interactive experience.

Users prefer AI’s personalized suggestions as they can enhance material searches and foster creativity. Through the

discussion, our participants emphasized the need for AI to provide visual enhancement suggestions based on their collage content. For example, they expected that AI could recommend matching colors, fonts, or layouts based on their selected materials and style, helping them quickly achieve their desired visual effect. We envision AI recommending relevant materials based on the content selected by users or their conversations with the system. For example, if a user selects materials related to “playground” and “students” or mentions topics related to school in a conversation, the AI system could suggest elements like “soccer ball” or “blackboard.” Such personalized suggestions can not only reduce the time spent searching for materials but also provide creative directions to help users overcome creative blocks [27].

6 Limitation and Future Work

There are several limitations to our study. *First*, the AI applications were not integrated into the digital collage tool, requiring users to switch interfaces, which may have affected their creation. *Second*, since all participants came from an academic environment and generally had high levels of AI usage experience and acceptance, this could have introduced bias in the study results. *Finally*, the small sample size and narrow age range of participants in this study may limit the generalizability of the findings. For future work, we plan to develop an AI-assisted digital collage tool based on the findings from our research and test it with more participants in real-life settings to evaluate its effectiveness.

7 Conclusion

This study explores the impact of AI on digital collage creation and user experiences through a design workshop with eight participants. Participants were tasked with creating their own digital collages using AI tools and later engaged in a focus group interview. The findings suggest that AI can act as a supportive assistant by improving creation efficiency, reducing barriers to entry, and offering creative inspiration, while also fostering positive experiences such as emotional resonance. However, participants expressed a need for greater autonomy, interactivity, and personalization in AI-assisted collage-making. Additionally, the study offers design implications for the development of future AI-powered digital collage tools.

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